The Fundamental Theorem of Arithmetic in Euclidean Domains

Lin Chen, Anika Sharma, Ellie Su



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1. Introduction

Prime numbers are the simplest nontrivial positive integers, yet together they form the fundamental building blocks of arithmetic. This idea is captured by the Fundamental Theorem of Arithmetic, which states that every integer greater than 1 can be expressed as a product of primes, uniquely up to ordering. The existence of greatest common divisors, modular arithmetic, and many algorithms in number theory all ultimately rely on this uniqueness.

At first glance, this may appear to be a special feature of the integers. However, many familiar properties of $\mathbb{Z}$ are consequences of more general algebraic structure. This raises the following question: which properties of the integers are responsible for unique factorization, and under what conditions can it be recovered in more general rings?

The answer is perhaps more intricate than it may seem. There are integral domains in which every nonzero element factors into irreducible terms, but such factorizations are not necessarily unique. An example is:

$$\mathbb{Z}\left[\sqrt{-5}\right] = \{a + b\sqrt{-5} \mid a, b \in \mathbb{Z}\},$$

where

$$6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5}).$$

All four factors appearing above are irreducible, and none differ from another by a unit. This shows that unique factorization is not a direct consequence of being an integral domain, but depends on additional structure.

One important class of rings in which this structure can be recovered is that of Euclidean domains. A Euclidean domain is an integral domain with a Euclidean norm, a function that assigns a measure of size to each nonzero element that is compatible with division. The key property is that for any a and $b \neq 0$, one can write

$$a = bq + r,$$

where either $r = 0$ or the remainder is strictly smaller than b with respect to the norm.

This condition yields the Euclidean Algorithm for computing greatest common divisor, which in turn produces Bézout's Identity, expressing a greatest common divisor as a linear combination of the original elements. From this, Euclid's Lemma follows: if an irreducible element divides a product, it must divide one of the factors. These results together form the mechanism behind unique factorization.

The purpose of this exposition is to prove the Fundamental Theorem of Arithmetic in Euclidean domains. We assume only the axioms of a commutative ring, the order properties of the natural numbers, and the Well-Ordering Principle. All subsequent results are derived entirely from these basic properties.

We begin by establishing the underlying algebraic framework: the ring axioms together with the order structure on the natural numbers and the Well-Ordering Principle, which serve as the only foundational assumptions throughout.

With this in place, we introduce the basic language of factorization in integral domains, including divisibility, units, irreducible elements, and prime elements, and we define Euclidean domains by their norm.

The Euclidean norm then provides a way to measure size, allowing arguments based on minimal elements and the Well-Ordering Principle. This leads to the Division Algorithm in Euclidean domains, which serves as the starting point for all further development. Once we have division with remainder, we construct the Euclidean algorithm and prove Bézout's Identity, showing that greatest common divisors can be written as linear combinations of the original elements.

The next step is Euclid's Lemma, which is key for defining uniqueness of factorization. This property is not guaranteed in arbitrary integral domains, but in a Euclidean domain it follows from Bézout's Identity, which itself is a consequence of the division structure provided by the Euclidean norm. We use Extended Euclidean Algorithm to compute the Bézout coefficients.

With these tools in place, the Fundamental Theorem of Arithmetic follows. Existence of factorization is obtained using the Well-Ordering Principle applied to the Euclidean norm, ensuring that every nonzero non-unit element eventually decomposes into irreducibles. Uniqueness follows from repeated applications of Euclid's Lemma, which forces any two factorizations to agree up to ordering and multiplication by units.

After that, we give some theoretical examples and real-world applications with the use of the Fundamental Theorem of Arithmetic.

The main result is that unique factorization is not an isolated property of the integers, but rather the outcome of a well-behaved notion of division captured by Euclidean domains.

2. Historical Background

Long before Euclidean domains had their official name, mathematicians were already chasing the mystery of factorization. In $\mathbb{Z}$, primes behave like tiny building blocks: every integer can be taken apart into primes, and no matter how you take it apart, the same primes appear in the end. Around 300 BCE, Euclid noticed the key trick behind this, now called Euclid's Lemma: if a prime divides a product, then it must divide one of the factors.

Later, mathematicians found that this same kind of magic happens in other worlds too. In polynomial rings over fields, we can divide with remainder, just like with integers. Euclidean domains capture this idea: they are rings where a norm keeps track of size, so division always produces a smaller remainder. From this division process come the Euclidean Algorithm, Bézout's Identity, Euclid's Lemma, and finally unique factorization.

But outside Euclidean domains, the spell can break. Some rings allow elements to factor in two genuinely different ways. So Euclidean domains are special because they are places where factorization stays orderly, even when the objects are no longer ordinary integers.The failure of unique factorization motivated much of nineteenth-century algebra.

3. Definitions

We now introduce the fundamental definitions used throughout the exposition. All definitions are made within a commutative ring R with identity $1 \neq 0$.

3.1. Basic Mathematical Notation

A **set** is a collection of objects called elements. If a is an element of a set A, we write

$$a \in A.$$

If a is not an element of A, we write

$$a \notin A.$$

Throughout this paper, letters such as R, A, and F will denote sets equipped with additional algebraic structure, while letters such as a, b, c will denote elements of those sets.

A **function** from a set A to a set B is a rule that assigns every element of A exactly one element of B. We write

$$f : A \to B.$$

If $a \in A$, then the element assigned to a is denoted by

$$f(a) \in B.$$

A function is therefore a mathematical object that associates elements of one set with elements of another set.

The natural numbers are defined by

$$\mathbb{N} = \{0, 1, 2, 3, \ldots\}.$$

The positive integers are

$$\mathbb{Z}^+ = \{1, 2, 3, \ldots\}.$$

The integers are

$$\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}.$$

The rational numbers are

$$\mathbb{Q} = \left\{ \frac{a}{b} : a, b \in \mathbb{Z},\ b \neq 0 \right\}.$$

The real numbers are denoted by $\mathbb{R}$.

3.2. Fields

Definition 1. A **field** is a set F equipped with two binary operations, addition and multiplication, satisfying the following properties:

1. Addition is associative:
$$(a + b) + c = a + (b + c).$$

2. Addition is commutative:
$$a + b = b + a.$$

3. There exists an additive identity $0 \in F$ such that
$$a + 0 = a$$
for every $a \in F$.

4. Every element $a \in F$ has an additive inverse $-a \in F$ satisfying

$$a + (-a) = 0.$$

5. Multiplication is associative:

$$(ab)c = a(bc).$$

6. Multiplication is commutative:

$$ab = ba.$$

7. There exists a multiplicative identity $1 \in F$, with $1 \neq 0$, such that

$$a \cdot 1 = a.$$

8. Every nonzero element $a \in F$ has a multiplicative inverse $a^{-1} \in F$ satisfying

$$aa^{-1} = 1.$$

9. Multiplication distributes over addition:

$$a(b + c) = ab + ac.$$

For example, $\mathbb{Q}$ is a field because every nonzero rational number has a multiplicative inverse.

Definition 2 (Negative of an Element)**.** Let F be a field and let $a \in F$. The **negative** of a, denoted by $-a$, is the unique element of F satisfying

$$a + (-a) = 0.$$

The element 0 is the additive identity of F.

The uniqueness of the additive inverse follows directly from the field axioms. Suppose $b, c \in F$ both satisfy

$$a + b = 0$$

and

$$a + c = 0.$$

Then

$$b = b + 0 = b + (a + c) = (b + a) + c = 0 + c = c.$$

Therefore,

$$b = c,$$

so the additive inverse is unique.

3.3. Rings

Definition 3. A **ring** is a set R equipped with two binary operations, addition and multiplication, satisfying the following axioms for all $a, b, c \in R$:

1. Addition is associative:
$$(a + b) + c = a + (b + c).$$

2. Addition is commutative:
$$a + b = b + a.$$

3. There exists an element $0 \in R$ such that
$$a + 0 = a.$$

4. For every $a \in R$, there exists an element $-a \in R$ such that
$$a + (-a) = 0.$$

5. Multiplication is associative:
$$(ab)c = a(bc).$$

6. Multiplication distributes over addition:
$$a(b + c) = ab + ac.$$

Definition 4. A ring R is called a **commutative ring with identity** if multiplication is commutative,

$$ab = ba,$$

and there exists an element $1 \in R$ with $1 \neq 0$ such that

$$a \cdot 1 = a$$

for every $a \in R$.

Definition 5. A **zero divisor** in a ring R is a nonzero element $a \in R$ for which there exists a nonzero element $b \in R$ satisfying

$$ab = 0.$$

Definition 6. A **commutative integral domain** is a commutative ring with identity that contains no zero divisors. Equivalently, if

$$ab = 0,$$

then

$$a = 0$$

or

$$b = 0.$$

3.4. Polynomial Rings

Definition 7. Let R be a ring. The **polynomial ring over** R, denoted by $R[x]$, is the set of all expressions of the form

$$a_0 + a_1x + a_2x^2 + \cdots + a_nx^n,$$

where each coefficient satisfies

$$a_i \in R$$

and n is a nonnegative integer.

Addition and multiplication of polynomials are defined using the addition and multiplication operations of the coefficients together with the distributive property.

For example,

$$\mathbb{Q}[x]$$

is the polynomial ring whose coefficients are rational numbers. An element of $\mathbb{Q}[x]$ has the form

$$a_0 + a_1 x + \cdots + a_n x^n,$$

where each coefficient $a_i \in \mathbb{Q}$.

3.5. Euclidean Domains

Definition 8. Let A be a commutative integral domain. A function

$$N : A \setminus \{0\} \to \mathbb{N}$$

is called a **Euclidean norm** if for every $a, b \in A$ with $b \neq 0$, there exist $q, r \in A$ such that

$$a = bq + r,$$

where either

$$r = 0$$

or

$$N(r) < N(b).$$

Definition 9. A **Euclidean domain** is a commutative integral domain equipped with a Euclidean norm.

Equivalently, a commutative integral domain A is a Euclidean domain if there exists a function

$$N : A \setminus \{0\} \to \mathbb{N}$$

such that for every $a, b \in A$ with $b \neq 0$, division with remainder is possible:

$$a = bq + r,$$

where either

$$r = 0$$

or

$$N(r) < N(b).$$

The Euclidean norm provides a notion of size that allows the Euclidean Algorithm to terminate after finitely many steps.

3.6. Divisibility

Definition 10. Let R be a ring and let $a, b \in R$. We say that a **divides** b, written

$$a \mid b,$$

if there exists some element $c \in R$ such that

$$b = ac.$$

The element c is called a quotient of b by a.

Definition 11. Two elements $a, b \in R$ are called **associates** if there exists a unit $u \in R$ such that

$$a = ub.$$

In other words, two elements are associates if they differ only by multiplication by an invertible element.

3.7. Units

Definition 12. A nonzero element $u \in R$ is called a **unit** if there exists an element $v \in R$ such that

$$uv = 1.$$

The element v is called the multiplicative inverse of u.

Thus, units are precisely the elements of a ring that are invertible under multiplication.

3.8. Greatest Common Divisors

Definition 13. Let R be an integral domain and let $a, b \in R$, not both zero. An element $d \in R$ is called a **greatest common divisor** of a and b if:

1. d divides both a and b:
$$d \mid a \quad \text{and} \quad d \mid b;$$

2. every common divisor of a and b divides d. That is, whenever

$$c \mid a \quad \text{and} \quad c \mid b,$$

then

$$c \mid d.$$

We write

$$d = \gcd(a, b).$$

3.9. Irreducible and Prime Elements

Definition 14. A nonzero element $p \in R$ that is not a unit is called **irreducible** if whenever

$$p = ab$$

for some $a, b \in R$, then either a or b is a unit.

Equivalently, an irreducible element cannot be written as a product of two nonunits.

Definition 15. A nonzero, nonunit element $p \in R$ is called **prime** if whenever

$$p \mid ab$$

for some $a, b \in R$, then

$$p \mid a$$

or

$$p \mid b.$$

3.10. Finite Sequences

Definition 16. A **finite sequence** of elements of a set S is an ordered list

$$(a_1, a_2, \ldots, a_n)$$

containing a finite number of elements of S.

The positive integer n is called the length of the sequence.

In the Euclidean Algorithm, the sequence of remainders is finite because the Euclidean norms form a strictly decreasing sequence of natural numbers.

3.11. Inequalities and Absolute Value

Definition 17. For integers a, b, we define

$$a < b \iff b - a \in \mathbb{Z}^+,$$

$$a > b \iff a - b \in \mathbb{Z}^+,$$

$$a \leq b \iff b - a \in \mathbb{N},$$

$$a \geq b \iff a - b \in \mathbb{N}.$$

Definition 18. For $x \in \mathbb{R}$, the **absolute value** of x, denoted by $|x|$, is defined as

$$|x| = \begin{cases} x, & \text{if } x \geq 0, \\ -x, & \text{if } x < 0. \end{cases}$$

3.12. Axioms

Having established the necessary algebraic definitions, we now specify the fundamental assumptions used throughout the remainder of the exposition. All results that follow will be derived from these axioms together with the definitions introduced previously.

The exposition relies on three main assumptions:

1. the axioms of a commutative ring with identity,
2. the standard ordering properties of the natural numbers,
3. the Well-Ordering Principle.

3.12.1. Ring Axioms

Throughout this paper, we work with commutative rings with identity. Therefore, we assume that the rings considered satisfy the following properties for all $a, b, c \in R$.

1. **Associativity of addition:**

$$(a + b) + c = a + (b + c).$$

2. **Commutativity of addition:**

$$a + b = b + a.$$

3. **Existence of an additive identity:**

   There exists an element $0 \in R$ such that

$$a + 0 = a.$$

4. **Existence of additive inverses:**

   For every $a \in R$, there exists an element $-a \in R$ such that

$$a + (-a) = 0.$$

5. **Associativity of multiplication:**

$$(ab)c = a(bc).$$

6. **Existence of a multiplicative identity:**

   There exists an element $1 \in R$ with

$$1 \neq 0$$

   such that

$$a \cdot 1 = a.$$

7. **Commutativity of multiplication:**

$$ab = ba.$$

8. **Distributive property:**

   Multiplication distributes over addition:

$$a(b + c) = ab + ac.$$

A structure satisfying these properties is called a commutative ring with identity.

3.12.2. Integral Domain Assumption

In addition to being a commutative ring with identity, the rings considered throughout this paper are assumed to be integral domains.

This means that the ring contains no zero divisors. In other words, for all $a, b \in R$,

$$ab = 0$$

implies that

$$a = 0$$

or

$$b = 0.$$

This assumption is important because many arguments involving factorization require that nonzero elements cannot multiply together to produce zero.

3.12.3. Axioms of the Natural Numbers

The Euclidean norm used in Euclidean domains takes values in the natural numbers, so we must specify the basic properties of the natural numbers and their ordering.

The natural numbers are

$$\mathbb{N} = \{0, 1, 2, 3, \ldots\},$$

together with the usual ordering relation $\leq$.

We assume that this ordering satisfies the following properties.

1. **The ordering is total.**

   For all $a, b, c \in \mathbb{N}$:

   - $a \leq a$,
   - if $a \leq b$ and $b \leq a$, then $a = b$,
   - if $a \leq b$ and $b \leq c$, then $a \leq c$,
   - either $a \leq b$ or $b \leq a$.

   Thus, any two natural numbers can always be compared using the ordering relation.

2. **Adding the same number preserves order.**

   If

$$a \leq b,$$

   then for every $c \in \mathbb{N}$,

$$a + c \leq b + c.$$

   Therefore, adding the same quantity to both sides of an inequality preserves the direction of the inequality.

3. **Multiplying by a natural number preserves order.**

   If

$$a \leq b,$$

   then for every $c \in \mathbb{N}$,

$$ac \leq bc.$$

   Therefore, multiplying both sides of an inequality by a natural number preserves the ordering.

3.12.4. Well-Ordering Principle

The final foundational assumption used throughout this paper is the Well-Ordering Principle.

[Well-Ordering Principle] Every nonempty subset of $\mathbb{N}$ contains a least element.

This principle allows us to select minimal elements from subsets of the natural numbers. It will later be used in arguments involving Euclidean norms, the termination of the Euclidean Algorithm, and the existence part of the Fundamental Theorem of Arithmetic.

4. Preliminaries

4.1. The Division Algorithm

Theorem 1. *Let A be a Euclidean domain with Euclidean norm N. For any $a, b \in A$ with $b \neq 0$, there exist $q, r \in A$ such that*

$$a = bq + r,$$

where either $r = 0$ *or* $N(r) < N(b)$.

Proof. Existence:

Consider the set $S = \{a - bk \mid k \in \mathbb{Z},\ a - bk \geq 0\}$. We claim that S is nonempty. If $a \geq 0$, then taking $k = 0$ gives $a - b \cdot 0 = a \geq 0$, so $a \in S$. If $a < 0$, taking $k = a$ gives $a - ba = a(1 - b)$. Since $a < 0$ and $1 - b \leq 0$, we have $a(1 - b) \geq 0$, so $a - ba \in S$. Therefore, S is nonempty.

By the Well-Ordering Principle, S has a least element. Let $r = a - bq$ be the least element of S for some $q \in \mathbb{Z}$. Since $r \in S$, we have $r \geq 0$.

We now show that $r < b$. Suppose, for contradiction, that $r \geq b$. Then $r-b \geq 0$. Moreover,$r-b = (a - bq) - b = a - b(q + 1)$. Thus $r - b \in S$. Since $b > 0$, we have $r - b < r$, which contradicts the fact that r is the least element of S. Therefore, $0 \leq r < b$.

Since $r = a - bq$, we obtain $a = bq + r$. Thus the required integers q and r exist.

Uniqueness: Suppose that $a = bq + r$ and $a = bq' + r'$, where $0 \leq r < b$ and $0 \leq r' < b$. Subtracting the two equations gives $b(q - q') = r' - r$.Since $0 \leq r, r' < b$, we have $-b < r' - r < b$. Hence $-b < b(q - q') < b$. Because $b > 0$, dividing by b gives $-1 < q - q' < 1$. Since $q - q' \in \mathbb{Z}$, it follows that $q - q' = 0$. Therefore, $q = q'$. Substituting this into $b(q - q') = r' - r$ gives $r = r'$. Thus q and r are unique. □

Example 1. *As an example, we consider the Euclidean Algorithm in the familiar setting of* $\mathbb{Z}$. *Suppose we wish to compute* $\gcd(252, 198)$.

$$252 = 198(1) + 54,$$

$$198 = 54(3) + 36,$$

$$54 = 36(1) + 18,$$

$$36 = 18(2) + 0.$$

Since the last nonzero remainder is 18, *we conclude that*

$$\gcd(252, 198) = 18.$$

4.2. The Euclidean Algorithm, the Bézout's Identity, and the Extended Euclidean Algorithem

Theorem 2. *Let* A *be a Euclidean domain with Euclidean norm* N. *For any* $a, b \in A$, *not both zero, repeated division produces a finite sequence of remainders whose last nonzero term is a greatest common divisor of* a *and* b.

Proof. Assume $b \neq 0$. We define a sequence of remainders r_i recursively. Let $r_{-1} = a$ and $r_0 = b$. For each integer $i \geq 1$, if $r_{i-1} \neq 0$, the division property guarantees the existence of $q_i, r_i \in A$ such that $r_{i-2} = r_{i-1}q_i + r_i$where either $r_i = 0$ or $N(r_i) < N(r_{i-1})$.

First, we prove that this process must terminate (i.e., there exists some index $m \geq 0$ such that $r_{m+1} = 0$). Let us define the set of norms of all active nonzero remainders: $S = \{N(r_i) \in \mathbb{N} \cup \{0\} \mid r_i \neq 0 \text{ for } i \geq 0\}$. If $r_i \neq 0$ for all $i \geq 0$, then the sequence of norms is strictly decreasing: for every $k \geq 0$, we have $N(r_{k+1}) < N(r_k)$. This implies that for any element $N(r_k) \in S$, there is a strictly smaller element $N(r_{k+1}) \in S$. Consequently, the set S has no least element. However, since S is a subset of the non-negative integers $\mathbb{N} \cup \{0\}$, this contradicts WOP. Therefore, the sequence of nonzero remainders cannot be infinite. There must exist a unique integer $n \geq 0$ such that $r_n \neq 0$ and $r_{n+1} = 0$.

Next, we prove that the last nonzero remainder r_n is a common divisor of a and b.

We want to show that $r_n \mid r_i$ for all $i \in I$. Let $S = \{i \in I \mid r_n \nmid r_i\}$. We want to show that S is empty. Suppose, for contradiction, that $S \neq \emptyset$. We define $T = \{n - i \mid i \in S\}$. Since $i \leq n$ for all $i \in S$, every element of T is non-negative, meaning $T \subseteq \mathbb{N} \cup \{0\}$. Because S is non-empty, T is a non-empty subset of non-negative integers.

By WOP, T contains a least element, which we denote by m. Let $i_0 = n - m \in S$ be the corresponding index. We analyze the possible values of m:

- If $m = 0$, then $i_0 = n$. Since $r_n \mid r_n$ trivially, $n \notin S$, which means $0 \notin T$, a contradiction.
- If $m = 1$, then $i_0 = n - 1$. The final division relation is $r_{n-1} = r_n q_{n+1} + r_{n+1}$. Since $r_{n+1} = 0$, we have $r_{n-1} = r_n q_{n+1}$, which implies $r_n \mid r_{n-1}$. Thus, $n - 1 \notin S$, meaning $1 \notin T$, a contradiction.

Therefore, we must have $m \geq 2$. Because m is the least element of T, the non-negative integers $m - 1$ and $m - 2$ cannot belong to T. This implies that their corresponding indices do not belong to S:

$$n - (m - 1) = i_0 + 1 \notin S \implies r_n \mid r_{i_0+1}$$

$$n - (m - 2) = i_0 + 2 \notin S \implies r_n \mid r_{i_0+2}$$

The division relation at step $i_0 + 2$ is: $r_{i_0} = r_{i_0+2} - r_{i_0+1}q_{i_0+2}$. Since r_n divides both terms on the right-hand side, it must divide r_{i_0}. This implies $i_0 \notin S$, which contradicts our assumption that $i_0 \in S$.

Thus, the set S must be empty, meaning $r_n \mid r_i$ for all $i \in I$. In particular, $r_n \mid r_{-1}$ (which is a) and $r_n \mid r_0$ (which is b). Hence, r_n is a common divisor of a and b.

Let $d \in A$ such that $d \mid a$ and $d \mid b$. We want to show that $d \mid r_i$ for all $i \in I$. Let $U = \{i \in I \mid d \nmid r_i\}$. We want to show that U is empty. Suppose for sake of contradiction that $U \neq \emptyset$. We define a set of non-negative integers V by shifting the indices:$V = \{i + 1 \mid i \in U\}$.

Since $i \geq -1$ for all $i \in I$, every element in V is non-negative, meaning $V \subseteq \mathbb{N} \cup \{0\}$. Because U is non-empty, V is a non-empty subset of non-negative integers.

By WOP, V must contain a least element, which we denote by p. Let $i_1 = p - 1 \in U$ be the corresponding index. We analyze the possible values of p:

- If $p = 0$, then $i_1 = -1$. But $r_{-1} = a$, and we assumed $d \mid a$, so $-1 \notin U$, meaning $0 \notin V$, a contradiction.
- If $p = 1$, then $i_1 = 0$. But $r_0 = b$, and we assumed $d \mid b$, so $0 \notin U$, meaning $1 \notin V$, a contradiction.

Therefore, we must have $p \geq 2$. Because p is the least element of V, the non-negative integers $p-1$ and $p-2$ cannot belong to V. This implies that their corresponding indices do not belong to U:

$$(p-1) - 1 = i_1 - 1 \notin U \implies d \mid r_{i_1 - 1}$$

$$(p-2) - 1 = i_1 - 2 \notin U \implies d \mid r_{i_1 - 2}$$

The division relation at step i_1 is:

$$r_{i_1} = r_{i_1 - 2} - r_{i_1 - 1} q_{i_1}$$

Since d divides both r_{i_1-2} and r_{i_1-1}, it must also divide r_{i_1}. This implies $i_1 \notin U$, which directly contradicts our assumption that $i_1 \in U$.

Thus, the set U must be empty, meaning $d \mid r_i$ for all $i \in I$. In particular, $d \mid r_n$.

Consequently, r_n satisfies both conditions of being a greatest common divisor of a and b. □

Theorem 3 (Bézout's Identity)**.** *Let A be a Euclidean domain. If $a, b \in A$ are not both zero and $d = \gcd(a, b)$, then there exist $x, y \in A$ such that*

$$d = ax + by.$$

Proof. By the Euclidean Algorithm, the greatest common divisor d is the last nonzero remainder r_n in the finite sequence of remainders indexed by $I = \{i \in \mathbb{Z} \mid -1 \leq i \leq n\}$ where $r_{-1} = a$ and $r_0 = b$.

We want to prove that every remainder r_i for $i \in I$ can be written as a linear combination of a and b. Let $U = \{i \in I \mid \text{there do not exist } x, y \in \Lambda \text{ such that } r_i = ax + by\}$. Suppose, for contradiction, that $U \neq \emptyset$.

We define a set of non-negative integers V by shifting the indices of $U : V = \{i + 1 \mid i \in U\}$. Since $i \geq -1$ for all $i \in I$, every element in V is non-negative, meaning $V \subseteq \mathbb{N} \cup \{0\}$. Because U is non-empty, V is a non-empty subset of non-negative integers.

By the WOP, V must contain a least element, which we denote by p. Let $i_0 = p - 1 \in U$ be the corresponding index in U. We analyze the possible values of p:

- If $p = 0$, then $i_0 = -1$. But $r_{-1} = a = a(1) + b(0)$, which is a linear combination of a and b. Thus, $-1 \notin U$, meaning $0 \notin V$, a contradiction.
- If $p = 1$, then $i_0 = 0$. But $r_0 = b = a(0) + b(1)$, which is a linear combination of a and b. Thus, $0 \notin U$, meaning $1 \notin V$, a contradiction.

Therefore, we must have $p \geq 2$. Because p is the least element of V, the non-negative integers $p-1$ and $p-2$ cannot belong to V. This implies that their corresponding indices do not belong to U:

$$(p-1) - 1 = i_0 - 1 \notin U \implies \exists\, x_{i_0-1}, y_{i_0-1} \in A \text{ such that } r_{i_0-1} = ax_{i_0-1} + by_{i_0-1}$$
$$(p-2) - 1 = i_0 - 2 \notin U \implies \exists\, x_{i_0-2}, y_{i_0-2} \in A \text{ such that } r_{i_0-2} = ax_{i_0-2} + by_{i_0-2}$$

The division relation at step i_0 is:

$$r_{i_0} = r_{i_0-2} - r_{i_0-1}q_{i_0}$$

Substituting the linear combinations for r_{i_0-2} and r_{i_0-1} yields:

$$\begin{aligned} r_{i_0} &= (ax_{i_0-2} + by_{i_0-2}) - (ax_{i_0-1} + by_{i_0-1})q_{i_0} \\ &= a(x_{i_0-2} - x_{i_0-1}q_{i_0}) + b(y_{i_0-2} - y_{i_0-1}q_{i_0}) \end{aligned}$$

Since A is a ring, $x_{i_0} = x_{i_0-2} - x_{i_0-1}q_{i_0} \in A$ and $y_{i_0} = y_{i_0-2} - y_{i_0-1}q_{i_0} \in A$. Thus:

$$r_{i_0} = ax_{i_0} + by_{i_0}$$

This shows that r_{i_0} can be written as a linear combination of a and b, which means $i_0 \notin U$. This contradicts our assumption that $i_0 \in U$.

By contradiction, the set U must be empty. Consequently, every element r_i for $i \in I$ can be written as a linear combination of a and b. In particular, for $i = n$, there exist $x_n, y_n \in A$ such that:

$$d = r_n = ax_n + by_n$$

□

Example 2. *Consider the previous Euclidean Algorithm computation for* $\gcd(252, 198) = 18$. *We work backward through the equations to express the greatest common divisor as a linear combination of* 252 *and* 198.

From

$$54 = 36(1) + 18,$$

we obtain

$$18 = 54 - 36.$$

Since

$$36 = 198 - 3(54),$$

substituting gives

$$18 = 54 - (198 - 3(54)) = 4(54) - 198.$$

Finally, since

$$54 = 252 - 198,$$

we have

$$18 = 4(252 - 198) - 198 = 4(252) - 5(198).$$

Thus,

$$18 = 4(252) - 5(198),$$

which is Bézout's Identity with coefficients $x = 4$ and $y = -5$. This motivates the Extended Euclidean Algorithm, which computes these coefficients directly.

Theorem 4 (Extended Euclidean Algorithm)**.** *Let A be a Euclidean domain with Euclidean function*

$$\delta : A \setminus \{0\} \to \mathbb{Z}_{\geq 0}.$$

For any $a, b \in A$, not both zero, the Euclidean Algorithm computes elements $x, y \in A$ such that

$$\gcd(a, b) = ax + by.$$

Proof. Without loss of generality, assume that $b \neq 0$. Set

$$r_{-1} = a \qquad \text{and} \qquad r_0 = b.$$

Whenever $r_{k-1} \neq 0$, the Euclidean property gives elements $q_k, r_k \in A$ such that

$$r_{k-2} = r_{k-1}q_k + r_k,$$

where either $r_k = 0$ or

$$\delta(r_k) < \delta(r_{k-1}).$$

We first prove that the Euclidean Algorithm terminates. Suppose, for contradiction, that it does not terminate. Then every remainder is nonzero, so the set

$$S = \{\delta(r_k) \mid k \in \mathbb{Z}_{\geq 0}\}$$

is a nonempty subset of $\mathbb{Z}_{\geq 0}$.

By the Well-Ordering Principle, S has a least element. Let

$$\delta(r_m) = \min S.$$

Since the algorithm does not terminate, $r_{m+1} \neq 0$. Therefore,

$$\delta(r_{m+1}) < \delta(r_m),$$

which contradicts the minimality of $\delta(r_m)$. Hence the algorithm must terminate.

Thus, for some $n \geq 0$,

$$r_n \neq 0 \qquad \text{and} \qquad r_{n+1} = 0.$$

We now prove that every remainder is a linear combination of a and b. The initial terms satisfy

$$r_{-1} = a = 1 \cdot a + 0 \cdot b$$

and

$$r_0 = b = 0 \cdot a + 1 \cdot b.$$

Suppose, for contradiction, that some remainder is not a linear combination of a and b. Consider the set

$$T = \{k \in \mathbb{Z}_{\geq 1} \mid r_k \text{ is not a linear combination of } a \text{ and } b\}.$$

By assumption, T is nonempty. By the Well-Ordering Principle, T has a least element; call it m.

Since m is the least element of T, both r_{m-2} and r_{m-1} are linear combinations of a and b. Thus there exist $s_{m-2}, t_{m-2}, s_{m-1}, t_{m-1} \in A$ such that

$$r_{m-2} = as_{m-2} + bt_{m-2}$$

and

$$r_{m-1} = as_{m-1} + bt_{m-1}.$$

From the Euclidean division equation

$$r_{m-2} = r_{m-1}q_m + r_m,$$

we obtain

$$r_m = r_{m-2} - r_{m-1}q_m.$$

Therefore,

$$\begin{aligned} r_m &= (as_{m-2} + bt_{m-2}) - (as_{m-1} + bt_{m-1})q_m \\ &= a(s_{m-2} - s_{m-1}q_m) + b(t_{m-2} - t_{m-1}q_m). \end{aligned}$$

Hence r_m is a linear combination of a and b, contradicting $m \in T$. Therefore, every remainder is a linear combination of a and b.

In particular, there exist $x, y \in A$ such that

$$r_n = ax + by.$$

We next prove that r_n is a greatest common divisor of a and b.

Let d be any common divisor of a and b. Suppose, for contradiction, that d does not divide every remainder. Consider

$$U = \{k \in \mathbb{Z}_{\geq 1} \mid d \nmid r_k\}.$$

If U is nonempty, then by the Well-Ordering Principle it has a least element, say m.

Since

$$d \mid r_{-1} = a \qquad \text{and} \qquad d \mid r_0 = b,$$

we have $m \geq 1$. By the minimality of m,

$$d \mid r_{m-2} \qquad \text{and} \qquad d \mid r_{m-1}.$$

But

$$r_m = r_{m-2} - r_{m-1}q_m,$$

so

$$d \mid r_m,$$

contradicting $m \in U$. Therefore, every common divisor of a and b divides every remainder. In

particular,

$$d \mid r_n.$$

It remains to show that r_n divides both a and b. Since

$$r_{n-1} = r_n q_{n+1} + r_{n+1}$$

and $r_{n+1} = 0$, we have

$$r_n \mid r_{n-1}.$$

Suppose, for contradiction, that r_n does not divide every earlier remainder. Consider

$$V = \{n - k \mid 0 \leq k \leq n + 1 \text{ and } r_n \nmid r_{k-1}\}.$$

If V is nonempty, then by the Well-Ordering Principle it has a least element. Equivalently, there is a largest index m such that

$$r_n \nmid r_{m-1}.$$

Since $r_n \mid r_n$ and $r_n \mid r_{n-1}$, we have $m \leq n - 1$.

By the maximality of m,

$$r_n \mid r_m \qquad \text{and} \qquad r_n \mid r_{m+1}.$$

From

$$r_{m-1} = r_m q_{m+1} + r_{m+1},$$

it follows that

$$r_n \mid r_{m-1},$$

a contradiction. Therefore,

$$r_n \mid r_{-1} = a \qquad \text{and} \qquad r_n \mid r_0 = b.$$

Thus r_n is a common divisor of a and b, and every common divisor of a and b divides r_n. Hence r_n is a greatest common divisor of a and b, up to multiplication by a unit.

Since

$$r_n = ax + by,$$

we conclude that

$$\gcd(a, b) = ax + by$$

after choosing the appropriate associate of r_n. □

4.3. Euclid's Lemma

Theorem 5. *In a Euclidean domain, every irreducible element is prime.*

Proof. Let $p \in A$ be irreducible, and suppose $p \mid ab$. We prove that $p \mid a$ or $p \mid b$. If $p \mid a$, we are done. Otherwise, suppose $p \nmid a$. Let $d = \gcd(p, a)$. Since $d \mid p$ and p is irreducible, d is either a unit or an associate of p. But if d were an associate of p, then $p \mid a$, contradicting $p \nmid a$. Hence d is a unit, so after multiplying by a unit we may write $\gcd(p, a) = 1$. By Bézout's Identity, there exist $x, y \in A$ such that

$$px + ay = 1.$$

Multiplying by b gives

$$pbx + aby = b.$$

Since $p \mid pbx$ and $p \mid aby$ because $p \mid ab$, it follows that $p \mid b$. Therefore p is prime. □

Example 3. *We first consider a familiar example in $\mathbb{Z}$. Since*

$$5 \mid 70$$

and

$$70 = 7 \cdot 10,$$

Euclid's Lemma implies that

$$5 \mid 7 \quad \textit{or} \quad 5 \mid 10.$$

Clearly,

$$5 \nmid 7,$$

so we conclude that

$$5 \mid 10.$$

This illustrates how primality forces divisibility of one of the factors.

Example 4. *We can also consider an example in the polynomial ring $\mathbb{Q}[x]$. Observe that*

$$(x + 1) \mid (x + 1)(x^2 + 1).$$

Since $x + 1$ is irreducible over $\mathbb{Q}$, Euclid's Lemma guarantees that whenever

$$(x+1) \mid fg,$$

then

$$(x+1) \mid f \quad or \quad (x+1) \mid g.$$

This is directly analogous to the integer case above.

5. The Fundamental Theorem of Arithmetic in Euclidean Domains

The Fundamental Theorem of Arithmetic in Euclidean domains states that every nonzero nonunit element in a Euclidean domain can be factored into irreducible elements, and that this factorization is unique up to reordering and multiplication by units.

5.1. Existence of Factorization

Lemma 6. *Every nonzero nonunit element of a Euclidean domain can be written as a product of irreducible elements.*

Proof. Let S be the set of non-zero non-unit elements of $\mathbb{A}$ that cannot be written as a product of irreducible elements. Suppose for contradiction that S is nonempty. By WOP, choose $a \in S$ such that $N(a)$ is minimal. Since a cannot be written as a product of irreducibles, a is not irreducible. Thus, $a = bc$ where $b, c \in \mathbb{A}$ are non-units. Since b and c are factors of a, we have $N(b) < N(a)$ and $N(c) < N(a)$. By minimality of a, both b and c can be written as products of irreducible elements. Therefore, $a = bc$ can also be written as a product of irreducible elements, contradicting $a \in S$. Hence, S is empty, so every non-zero non-unit element of $\mathbb{A}$ can be written as a product of irreducible elements. □

Example 5. *In $\mathbb{Z}$, consider the integer*

$$360 = 2^3 \cdot 3^2 \cdot 5.$$

Each factor is irreducible, so this is a factorization into irreducible elements.
Similarly, in $\mathbb{Q}[x]$,

$$x^4 - 1 = (x-1)(x+1)(x^2+1),$$

which is a factorization into irreducible polynomials over $\mathbb{Q}$.

5.2. Uniqueness of Factorization

Theorem 7 (Fundamental Theorem of Arithmetic for Euclidean Domains)**.** *Let A be a Euclidean domain. Every nonzero nonunit element of A can be written as a product of irreducible elements. Moreover, this factorization is unique up to reordering and multiplication by units.*

Proof. Existence was proved in the previous lemma. For uniqueness, suppose

$$f = p_1 p_2 \cdots p_m = q_1 q_2 \cdots q_n,$$

where all p_i and q_j are irreducible. Since $p_1 \mid q_1 q_2 \cdots q_n$ and every irreducible element in a Euclidean domain is prime, Euclid's Lemma implies that $p_1 \mid q_j$ for some j. Since q_j is irreducible, p_1 and q_j are associates. After reordering the q_j terms, assume p_1 is associate to q_1. Then $q_1 = u_1 p_1$ for some unit u_1. Substituting this into the original equality gives $p_1 p_2 \cdots p_m = u_1 p_1 q_2 \cdots q_n$. Since A is an integral domain and $p_1 \neq 0$, we can cancel p_1 to obtain $p_2 \cdots p_m = u_1 q_2 \cdots q_n$.

Since u_1 is a unit, $p_2 \mid q_2 \cdots q_n$. Repeating the same argument, after reordering the remaining factors, we have $q_2 = u_2 p_2$ for some unit u_2. Substituting and canceling p_2 gives $p_3 \cdots p_m = u_1 u_2 q_3 \cdots q_n$. Continuing in this way, each p_i is associate to a distinct q_j.

If $m < n$, then after canceling all the p_i, we obtain $1 = u_1 u_2 \cdots u_m q_{m+1} \cdots q_n$. This would imply that $q_{m+1} \cdots q_n$ is a unit, which is impossible because each q_j is irreducible and therefore a nonunit. Similarly, $n < m$ is impossible. Therefore $m = n$, and after reordering, each p_i differs from the corresponding q_i by a unit. Thus the factorization is unique up to reordering and multiplication by units. □

5.3. Fundamental Theorem of Arithmetic Outside of Euclidean Domains

The Fundamental Theorem of Arithmetic depends on the ability to control division. In Euclidean domains, the norm lets us repeatedly divide and produce smaller remainders. This gives the Euclidean Algorithm, Bézout's Identity, Euclid's Lemma, and finally unique factorization.

Outside Euclidean domains, the theorem can fail. For example, in $\mathbb{Z}[\sqrt{-5}]$,

$$6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5}),$$

and these are different factorizations into irreducible elements. Thus unique factorization is not automatic in every integral domain.

6. Examples

6.1. $\mathbb{Z}$ is a Euclidean Domain

Proof. Define $N : \mathbb{Z} \setminus \{0\} \to \mathbb{N}$ by $N(n) = |n|$. Let $a, b \in \mathbb{Z}$ with $b \neq 0$. By the ordinary division algorithm, there exist $q, r \in \mathbb{Z}$ such that

$$a = bq + r$$

and $0 \leq r < |b|$. Hence either $r = 0$ or

$$N(r) = |r| < |b| = N(b).$$

Since $\mathbb{Z}$ is an integral domain and satisfies the Euclidean division property, $\mathbb{Z}$ is a Euclidean domain. □

6.2. Polynomial Rings Over Fields

Let F be a field. Then $F[x]$ is a Euclidean domain with norm given by degree:

$$N(f) = \deg(f).$$

For any $f, g \in F[x]$ with $g \neq 0$, polynomial long division gives

$$f = gq + r,$$

where either $r = 0$ or $\deg(r) < \deg(g)$. Therefore $F[x]$ is a Euclidean domain. This example explains why polynomial factorization over a field behaves much like prime factorization in $\mathbb{Z}$.

7. Applications

7.1. Example 1: QR Codes and Reed–Solomon Error Correction

One important application of the Fundamental Theorem of Arithmetic for Euclidean domains is the error-correcting mechanism used in QR codes. Quick Response (QR) codes are two-dimensional matrix barcodes invented in 1994 by the Japanese company Denso Wave. Unlike traditional one-dimensional barcodes, QR codes can store a much larger amount of information, including text, URLs, contact information, and payment data. Their square grid structure allows data to be scanned rapidly from any direction, making them widely used in mobile payments, product tracking,

ticketing, authentication, and inventory management. A key feature of QR codes is their built-in error correction, which enables accurate decoding even when part of the code is damaged or obscured. QR codes remain readable even when part of the symbol is damaged because they employ Reed–Solomon codes, which perform polynomial arithmetic over the Euclidean domain.

Encoding Process The information stored in a QR code is first converted into a binary sequence. For example,

`HELLO`

may be represented as

$$01001010\cdots.$$

The binary data are then interpreted as the coefficients of a polynomial

$$m(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n,$$

where

$$a_i \in GF(256).$$

Since

$$256 = 2^8,$$

there exists a finite field with exactly 256 elements. A common construction is

$$GF(256) \cong GF(2)[x]/(f(x)),$$

where $f(x)$ is an irreducible polynomial of degree 8. Because the coefficients belong to a field, the polynomial ring

$$GF(256)[x]$$

is a Euclidean domain.

Instead of transmitting $m(x)$ directly, Reed–Solomon encoding computes

$$m(x)x^r,$$

where r is the number of error-correction symbols, and divides it by a generator polynomial $g(x)$:

$$m(x)x^r = q(x)g(x) + r(x).$$

The transmitted codeword is

$$c(x) = m(x)x^r + r(x).$$

This encoding process relies on polynomial division, which is possible because $GF(256)[x]$ is a Euclidean domain.

Decoding Process Suppose that part of the QR code is damaged during transmission. For example,

$$101011001\cdots$$

may be received as

$$101011101\cdots.$$

The decoder first computes the syndrome polynomial. If transmission errors are detected, it constructs the error locator polynomial and the error evaluator polynomial.

These polynomials are obtained using the Extended Euclidean Algorithm over the Euclidean domain $GF(256)[x]$. Since Bézout's Identity holds in every Euclidean domain, the algorithm computes the Bézout coefficients together with the greatest common divisor. This allows the decoder to determine the locations and magnitudes of transmission errors efficiently.

Once the error locations are known, the corrupted symbols are corrected and the original message is recovered.

Connection to the Fundamental Theorem of Arithmetic The Fundamental Theorem of Arithmetic for Euclidean domains states that every nonzero nonunit polynomial in $GF(256)[x]$ factors uniquely into irreducible polynomials (up to multiplication by units and ordering of the factors).

This unique factorization guarantees that irreducible polynomial decompositions and greatest common divisors are uniquely determined. Consequently, the algebraic objects used during Reed–Solomon decoding are mathematically well-defined and unique. While the Extended Euclidean Algorithm performs the actual computations, the Fundamental Theorem of Arithmetic provides the theoretical guarantee that these computations lead to a unique decoding result.

Therefore, the Euclidean domain structure supplies the computational tools (polynomial division, Bézout's Identity, and the Extended Euclidean Algorithm), whereas the Fundamental Theorem of Arithmetic provides the uniqueness that ensures the correctness and reliability of QR-code error correction.

7.2. Example 2: Cryptography and Unique Factorization

Another important application of the Fundamental Theorem of Arithmetic for Euclidean domains appears in cryptography. Many cryptographic systems rely on the idea that factorization is unique but computationally difficult to reverse.

Mathematical Principle In the integer ring $\mathbb{Z}$, which is a Euclidean domain, the classical Fundamental Theorem of Arithmetic states that every integer greater than 1 can be written uniquely as a product of prime numbers. More generally, in any Euclidean domain, every nonzero nonunit element has a unique factorization into irreducible elements, up to multiplication by units and reordering.

In RSA cryptography, the system chooses two large prime numbers

$$p \text{ and } q,$$

and multiplies them to form

$$N = pq.$$

The number N is made public, but the primes p and q are kept secret.

The Role of FTA The role of the Fundamental Theorem of Arithmetic is to guarantee that the factorization of N is unique:

$$N = pq.$$

This means that there is only one correct prime decomposition of N, up to order. Therefore, recovering p and q from N is a well-defined problem. If factorization were not unique, then the structure of the cryptographic system would become ambiguous, since the same public modulus could correspond to different private decompositions.

Thus, the FTA provides the algebraic foundation for the security assumption: although the prime factorization is unique, finding it for a very large N is computationally difficult.

The Role of the Euclidean Algorithm, Bézout's Identity, and the Extended Euclidean Algorithm The Euclidean Algorithm is first used to compute

$$\gcd(e, \varphi(N)).$$

If the gcd is equal to 1, then Bézout's Identity guarantees the existence of integers d and y satisfying

$$ed + \varphi(N)y = 1.$$

The Extended Euclidean Algorithm is then applied to compute these Bézout coefficients explicitly, yielding the integer d, which satisfies

$$ed \equiv 1 \pmod{\varphi(N)}.$$

The number d becomes the private decryption key.

Conclusion Therefore, in cryptography, the Fundamental Theorem of Arithmetic guarantees the uniqueness of prime factorization, while the Euclidean Algorithm efficiently computes greatest common divisors. Bézout's Identity guarantees the existence of modular inverses whenever the greatest common divisor is 1, and the Extended Euclidean Algorithm computes these inverses explicitly. Together, these results provide both the theoretical foundation and the computational tools underlying modern cryptographic systems and error-correcting codes.

7.3. Example 3: Signal Processing

A second application appears in signal processing, digital communication, and fast computation. In areas such as 5G communication, radar image processing, and fast Fourier transform algorithms, large computations are often broken into smaller independent computations and then recombined.

Euclidean domain The arithmetic used in signal processing depends on the underlying algebraic structure. In applications involving integer arithmetic, computations are performed in the Euclidean domain $\mathbb{Z}$.

In applications involving error-correcting codes or polynomial transforms, computations are carried out in the polynomial ring $GF(q)[x]$, which is also a Euclidean domain because $GF(q)$ is a field.

Mathematical Principle Suppose a system needs to perform computations modulo a large number M. If

$$M = m_1 m_2 \cdots m_k,$$

where $m_1, m_2, \ldots, m_k$ are pairwise coprime, then the Chinese Remainder Theorem states that

$$\mathbb{Z}/M\mathbb{Z} \cong \mathbb{Z}/m_1\mathbb{Z} \times \mathbb{Z}/m_2\mathbb{Z} \times \cdots \times \mathbb{Z}/m_k\mathbb{Z}.$$

This means that a large computation modulo M can be replaced by several smaller computations modulo $m_1, m_2, \ldots, m_k$.

The Role of FTA The Fundamental Theorem of Arithmetic guarantees that the modulus M has a unique prime factorization:

$$M = p_1^{a_1} p_2^{a_2} \cdots p_r^{a_r}.$$

This unique factorization allows us to decompose M into pairwise coprime components. For example,

$$M = m_1 m_2 \cdots m_k,$$

where

$$\gcd(m_i, m_j) = 1 \text{ for } i \neq j.$$

Because the decomposition is unique, the smaller modular systems are mathematically well-defined. This is what allows one large computation to be divided into several independent smaller computations without ambiguity.

Parallel Computation A number X modulo M can be represented by its residues:

$$X \mapsto (x_1, x_2, \ldots, x_k),$$

where

$$x_i \equiv X \pmod{m_i}.$$

Each smaller computation can be processed separately. This is useful in signal processing and hardware computation because different parts of the calculation can be performed in parallel.

The Role of the Euclidean Algorithm, Bézout's Identity, and the Extended Euclidean Algorithm After the smaller computations are finished, the system must reconstruct the original

result modulo M. For this, we define

$$M_i = \frac{M}{m_i}.$$

To recombine the residues, we need the modular inverse of M_i modulo m_i:

$$M_i M_i^{-1} \equiv 1 \pmod{m_i}.$$

The Euclidean Algorithm is first used to verify that

$$\gcd(M_i, m_i) = 1.$$

Since the greatest common divisor is 1, Bézout's Identity guarantees the existence of integers u_i and v_i satisfying

$$M_i u_i + m_i v_i = 1.$$

The Extended Euclidean Algorithm is then used to compute these Bézout coefficients explicitly. Hence,

$$u_i \equiv M_i^{-1} \pmod{m_i}.$$

The final reconstruction formula is

$$X \equiv \sum_{i=1}^{k} x_i M_i u_i \pmod{M}.$$

Conclusion Therefore, the Fundamental Theorem of Arithmetic guarantees the unique decomposition of the modulus into coprime components. The Euclidean Algorithm verifies that the required moduli are relatively prime, Bézout's Identity guarantees the existence of the corresponding modular inverses, and the Extended Euclidean Algorithm computes these inverses efficiently. Together, these results enable the Chinese Remainder Theorem to reconstruct the original solution from its smaller congruences, supporting fast and reliable parallel computation in signal processing.

7.4. Logic Map

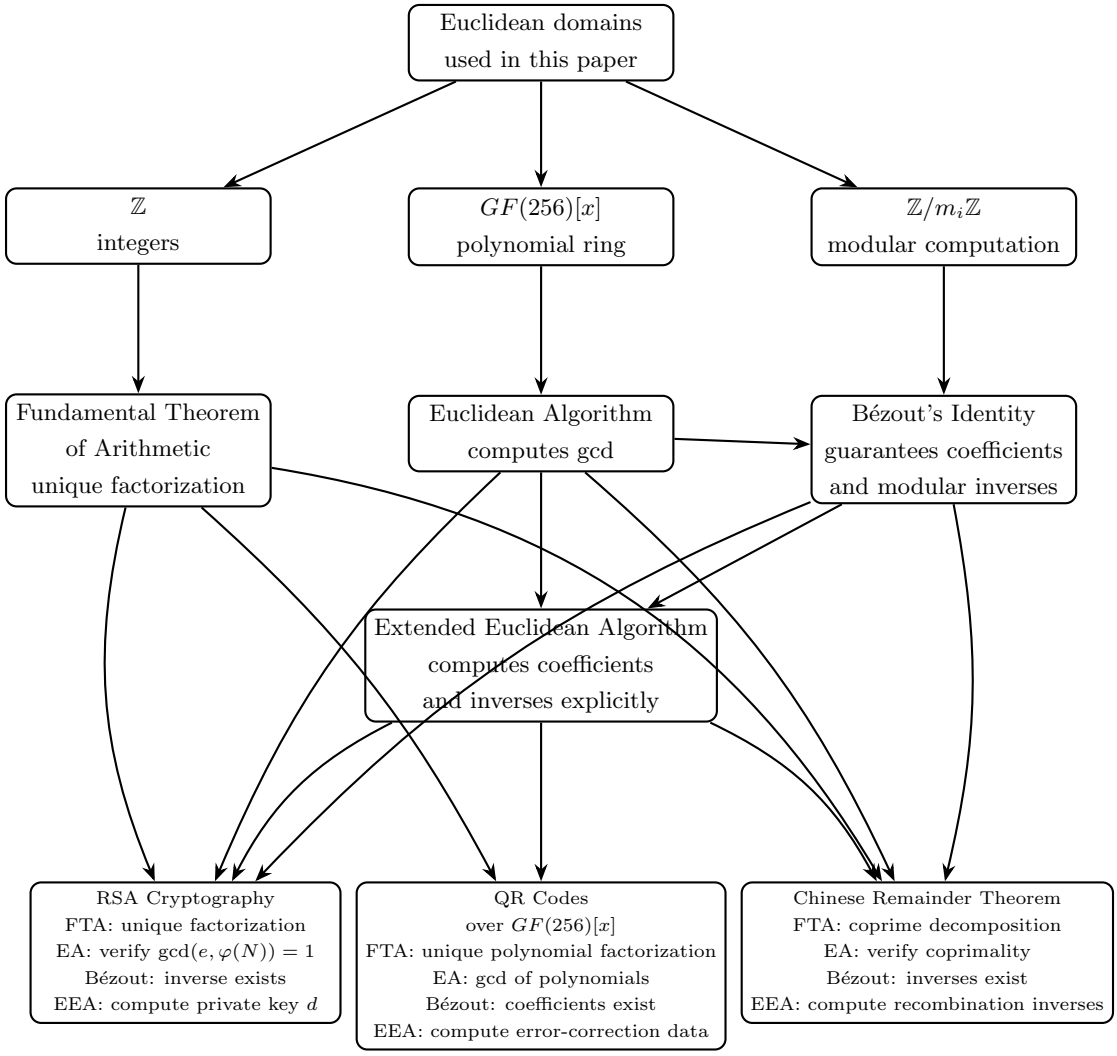


Figure 1: Logic map showing the Euclidean domains used in this paper and the roles of the Fundamental Theorem of Arithmetic, the Euclidean Algorithm, Bézout's Identity, and the Extended Euclidean Algorithm in each application.

8. Conclusion

Euclidean domains generalize the familiar arithmetic of the integers. The key structure is division with remainder: once a ring has a norm that makes every remainder smaller than the divisor, the Euclidean Algorithm follows. From the Euclidean Algorithm we obtain Bézout's Identity, and from Bézout's Identity we prove Euclid's Lemma. These steps are the machinery behind unique factorization.

Thus the Fundamental Theorem of Arithmetic is not just a fact about ordinary integers. It is a theorem about any algebraic world where division behaves well enough. In Euclidean domains, factorization remains orderly: every nonzero nonunit element breaks into irreducible pieces, and those pieces are unique up to order and units.

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